

Relation Algebra

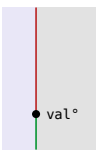
§6. Recursive Programs

§6.1. Digits of a number

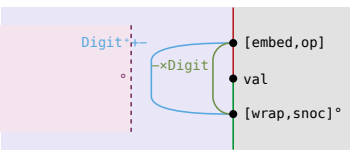
(6.1a)

$$\text{val}^\circ = \text{embed}^\circ \text{wrap} \text{ u } \text{op}^\circ - \times \text{Digit}(\text{val}^\circ) \text{snoc}$$

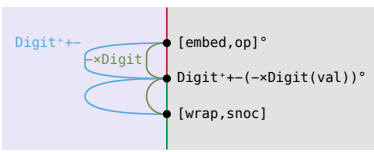
a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest



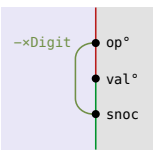
(a)



(b)



(c)



(d)



(a) $\text{val} = ([\text{embed}, \text{op}])$ — definition

(b) $(\varphi) = \alpha^\circ F((\varphi)) \varphi$ — catamorphisms

(c) $F(R)^\circ = F(R^\circ)$ — $F = (-\times \text{Digit})(\text{Digit}^{++})$, definition of F

(d) $\alpha = [\text{wrap}, \text{snoc}]$, then $(1+S)[P, Q] = [P, SQ]$ and $[g, h]^\circ [P, Q] = g^\circ \text{Puh}^\circ Q$ — coproduct

§6.2. $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

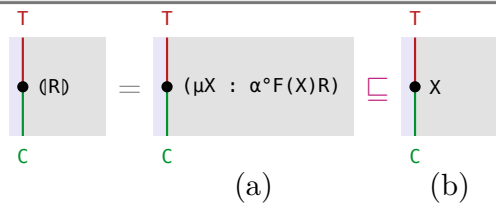
(6.2b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
$\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation

$$\alpha^\circ F(X) R \sqsubseteq X \Rightarrow (R) \sqsubseteq X$$

(6.2c)

a relation X that contains one unfolding of the fold's recursion at X contains the fold



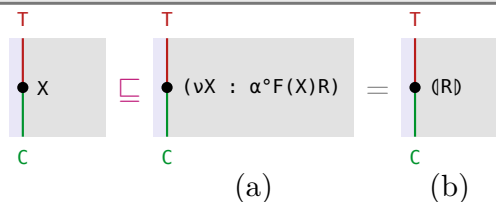
(a) (R) the least fixed point — Theorem 6.1, (6.2b)

(b) $\alpha^\circ F(X) R \sqsubseteq X$, μ below every prefix point

$$X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq (R)$$

(6.2d)

a relation X contained in one unfolding of the fold's recursion at X is contained in the fold



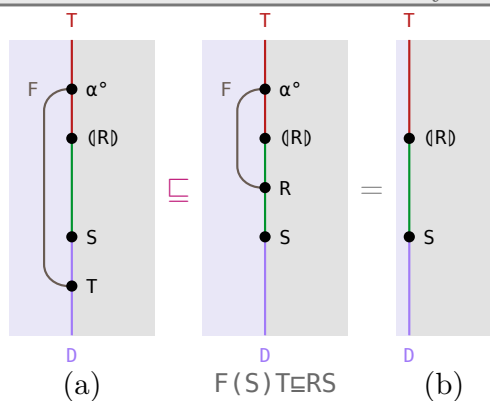
(a) $X \sqsubseteq \alpha^\circ F(X) R$, ν above every postfix point

(b) (R) the greatest fixed point

$$F(S) T \sqsubseteq R \Rightarrow (T) \sqsubseteq (R) S$$

(6.2e)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S



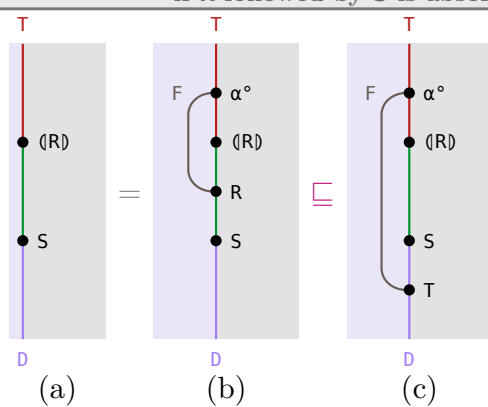
(a) (6.2) at $X \triangleq (R) S$

$F(RS) = F(R) F(S)$ — $\triangleq$ relator

(b) $F((R)) R = \alpha(R)$, $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce

$$RS \sqsubseteq F(S)T \Rightarrow (R)S \sqsubseteq (T)$$

(6.2f)

 if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T


- (a) (6.3) at $X \triangleq (R)S$
 (b) $\alpha(R) = F((R))R$, $\alpha^\circ \alpha = 1$ — reduce
 (c) $RS \sqsubseteq F(S)T$
 $F(R)F(S) = F(RS)$ — relator

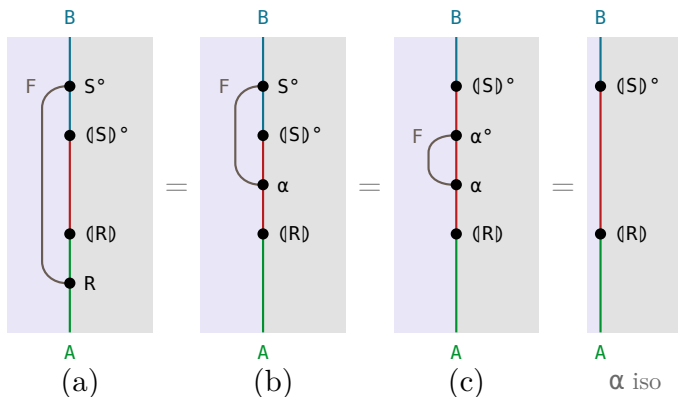
§6.3. $(S)^\circ (R) = (\mu X : S^\circ F(X)R)$

$$(\mathbb{S})^\circ(\mathbb{R}) = (\mu X : \mathbb{S}^\circ F(X)R)$$

(6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

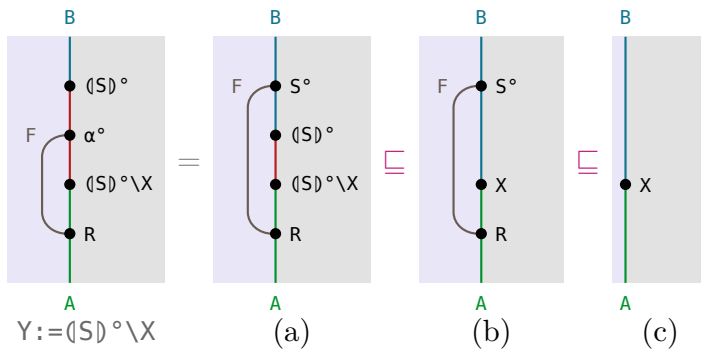
$$\mathbb{S}^\circ F((\mathbb{S})^\circ(\mathbb{R}))R = (\mathbb{S})^\circ(\mathbb{R})$$

 hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term $\mathbb{S}^\circ$ represents the decomposition stage, $F((\mathbb{S})^\circ(\mathbb{R}))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $S : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial

 (a) the body at $(\mathbb{S})^\circ(\mathbb{R})$
 $F(RS) = F(R)F(S) \triangleq$ relator

 (b) $F((\mathbb{R}))R = \alpha(\mathbb{R}) \triangleq$ reduce

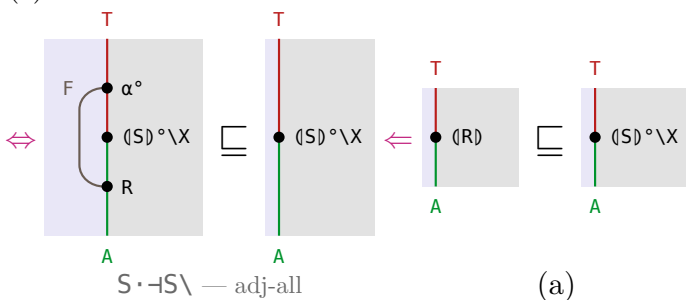
 (c) $(\mathbb{S})^\circ \alpha^\circ = \mathbb{S}^\circ F((\mathbb{S})^\circ) \triangleq$ reduce, relator-laws

$$\mathbb{S}^\circ F(X)R \sqsubseteq X \implies (\mathbb{S})^\circ(\mathbb{R}) \sqsubseteq X$$

 hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(\mathbb{S})^\circ(\mathbb{R})$ is included in X if X satisfies the associated recursion inequation

 $Y := (\mathbb{S})^\circ \setminus X$

 (a) $(\mathbb{S})^\circ \alpha^\circ = \mathbb{S}^\circ F((\mathbb{S})^\circ)$
 $F(RS) = F(R)F(S) \triangleq$ relator

 (b) $(\mathbb{S})^\circ((\mathbb{S})^\circ \setminus X) \sqsubseteq X \triangleq$ adj-all

 (c) $\mathbb{S}^\circ F(X)R \sqsubseteq X$

 $S \cdot \setminus S \triangleq$ adj-all

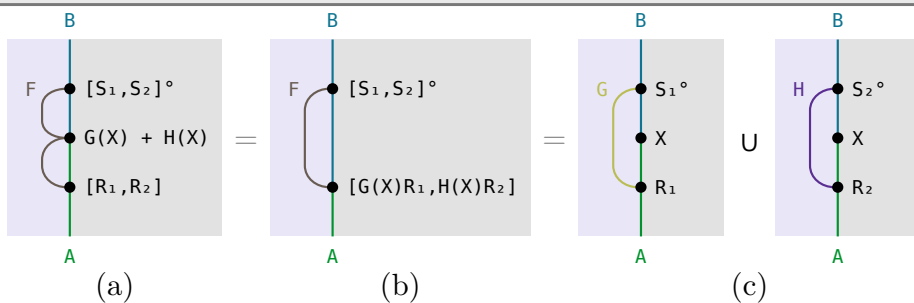
(a)

 (a) $(\mathbb{R}) = (\mu X : \alpha^\circ F(X)R) \triangleq$ reduce, (6.2b), adj-all

$$(\forall x. \forall y. \forall X. F(X) = G(X) + H(X)) \implies ([S_1, S_2])^\circ([R_1, R_2]) = (\mu X : S_1^\circ G(X)R_1 \cup S_2^\circ H(X)R_2)$$

(6.3b)

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results


 (a) the body of Theorem 6.2 at $F(X) = G(X) + H(X)$

 (b) $(P+Q)[R_1, R_2] = [PR_1, QR_2] \triangleq$ coproduct

 (c) $[S_1, S_2]^\circ[P, Q] = S_1^\circ P \cup S_2^\circ Q \triangleq$ coproduct

§6.4. Fast exponentiation and modulus computation

§6.5. Unique fixed points

§6.6. Sorting by selection

§6.7. Closure